Confusion Matrix –Decision Tree

```
[[76  9]
 [ 8 41]]
```

```
print(clf_report)
```

	precision	recall	f1-score	support
0	0.90	0.89	0.90	85
1	0.82	0.84	0.83	49
accuracy			0.87	134
macro avg	0.86	0.87	0.86	134
weighted avg	0.87	0.87	0.87	134

1. How is Precision calculated for Class 1?

Solution:

Formula:

Precision=TP/(TP+FP)

TP = 41

FP=9

41/50=0.82

2. How is Recall calculated for Class 1?

Solution:

Formula:

Recall=TP/(TP+FN)

TP=41

FN=8

Recall = 0.84

3. How is F1-measure calculated?

Solution:

Formula:

F1=2·(Precision)/(Recall+Precision)

=2*0.82/(0.82+0.84)

=0.83

4. Can you derive Accuracy from this report?

Solution:

Formula:

Accuracy=(TP+TN)/Total

TP+TN=76+41=117

Total=117+8+9=134

Accuracy= 0.87

5. What is Specificity?

Solution:

Formula:

Specificity=TN/(TN+FP)

=76 / (76 + 9) ≈ 0.89

6. Can you reconstruct the Confusion Matrix (approx)?

Solution:

From support:

- Class 0 → 85
- Class 1 → 49

From recall:

- Class 1 recall = 0.84 → TP ≈ 41, FN ≈ 8
- Class 0 recall = 0.89 → TN ≈ 76, FP ≈ 9

	Pred 0	**Pred 1**
Actual 0	76	9
Actual 1	8	41

7. How is Macro Average calculated-precision?

Solution:

Formula:

Macro Average= (P(1)+P(0))/2

=(0.9+0.83)/2

Macro Average = 0.865

8. How is Weighted Average-Precision calculated?

Solution:

Formula:

WA = P(0) *Support (0) + P(1)* Support(1)/ Total

= 0.90 × 85 + 0.83 × 49) / 134 ≈ 0.87

9. How do precision and recall relate mathematically?

Solution:

- Precision depends on **false positives**
- Recall depends on **false negatives**